

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
```

First we perform EDA

```
data = pd.read_csv('/content/house_rent.csv')
data.head()
```

	Posted On	BHK	Rent	Size	Floor	Area Type	Area Locality	City	Furnishing Status	Tenant Preferred	Bathroom	Point of Contact
0	2022-05-18	2	10000	1100	Ground out of 2	Super Area	Bandel	Kolkata	Unfurnished	Bachelors/Family	2	Contact Owner
1	2022-05-13	2	20000	800	1 out of 3	Super Area	Phool Bagan, Kankurgachi	Kolkata	Semi-Furnished	Bachelors/Family	1	Contact Owner
2	2022-05-16	2	17000	1000	1 out of 3	Super Area	Salt Lake City Sector 2	Kolkata	Semi-Furnished	Bachelors/Family	1	Contact Owner
3	2022-07-04	2	10000	800	1 out of 2	Super Area	Dumdum Park	Kolkata	Unfurnished	Bachelors/Family	1	Contact Owner
4	2022-05-09	2	7500	850	1 out of 2	Carpet Area	South Dum Dum	Kolkata	Unfurnished	Bachelors	1	Contact Owner

```
data.value_counts()
```

													count
Posted On	BHK	Rent	Size	Floor	Area Type	Area	Locality	City	Furnishing Status	Tenant Preferred	Bathroom	Point of Contact	
2022-07-11	2	12000	550	Ground out of 1	Super Area		Choolaimedu	Chennai	Unfurnished	Bachelors/Family	2	Contact Owner	1
2022-04-13	3	260000	1800	10 out of 11	Carpet Area		JVPD Scheme	Mumbai	Furnished	Family	4	Contact Agent	1
2022-04-23	1	6500	800	Ground out of 1	Carpet Area		Santragachi	Kolkata	Unfurnished	Bachelors/Family	1	Contact Owner	1
		15000	1000	2 out of 3	Super Area		Barrackpore	Kolkata	Unfurnished	Bachelors/Family	1	Contact Owner	1
	2	7000	450	1 out of 3	Carpet Area		safuipara	Kolkata	Furnished	Bachelors	2	Contact Owner	1
...	...	...	...	...	...		...	...	...	...	...	...	...
2022-04-25	1	6000	220	Ground out of 2	Super Area		Bidhan Pally, Bansdroni	Kolkata	Furnished	Bachelors/Family	1	Contact Owner	1
2022-04-24	3	15000	1350	1 out of 2	Carpet Area		Annapurna Apartment, Rajarhat Chowmatha	Kolkata	Unfurnished	Family	2	Contact Owner	1
		12000	1000	Ground out of 1	Super Area		Khardah	Kolkata	Unfurnished	Bachelors/Family	2	Contact Owner	1
	2	11000	1000	Ground out of 1	Super Area		Anandapuri, Barrackpore	Kolkata	Furnished	Bachelors/Family	2	Contact Owner	1
		9000	650	2 out of 3	Carpet Area		tollygunge Kabardanga	Kolkata	Unfurnished	Bachelors/Family	1	Contact Owner	1

4746 rows × 1 columns

dtype: int64

data.describe()

	BHK	Rent	Size	Bathroom
count	4746.000000	4.746000e+03	4746.000000	4746.000000
mean	2.083860	3.499345e+04	967.490729	1.965866
std	0.832256	7.810641e+04	634.202328	0.884532
min	1.000000	1.200000e+03	10.000000	1.000000
25%	2.000000	1.000000e+04	550.000000	1.000000
50%	2.000000	1.600000e+04	850.000000	2.000000
75%	3.000000	3.300000e+04	1200.000000	2.000000
max	6.000000	3.500000e+06	8000.000000	10.000000

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4746 entries, 0 to 4745
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Posted On             4746 non-null   object
1   BHK                   4746 non-null   int64
2   Rent                  4746 non-null   int64
3   Size                  4746 non-null   int64
4   Floor                 4746 non-null   object
5   Area Type             4746 non-null   object
6   Area Locality         4746 non-null   object
7   City                  4746 non-null   object
8   Furnishing Status     4746 non-null   object
9   Tenant Preferred      4746 non-null   object
10  Bathroom              4746 non-null   int64
11  Point of Contact      4746 non-null   object
dtypes: int64(4), object(8)
memory usage: 445.1+ KB
```

```
# Check the data for nulls
data.isnull().sum()
```

	0
Posted On	0
BHK	0
Rent	0
Size	0
Floor	0
Area Type	0
Area Locality	0
City	0
Furnishing Status	0
Tenant Preferred	0
Bathroom	0
Point of Contact	0

dtype: int64

```
# unique values of city  
data['City'].unique()
```

```
array(['Kolkata', 'Mumbai', 'Bangalore', 'Delhi', 'Chennai', 'Hyderabad'],  
      dtype=object)
```

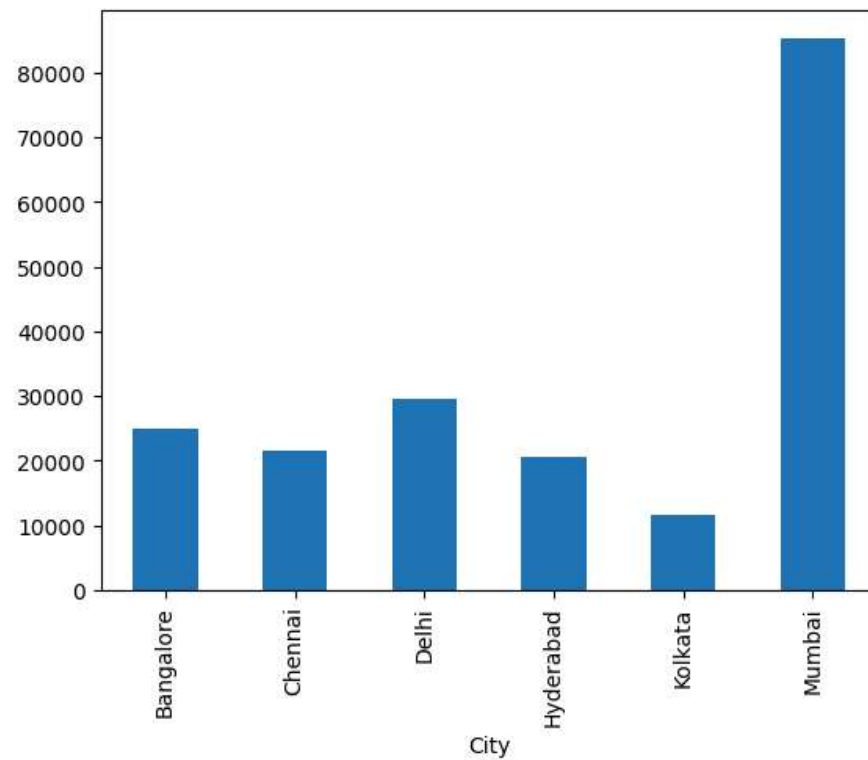
```
# average rent with city  
data.groupby('City')['Rent'].mean()
```

	Rent
City	
Bangalore	24966.365688
Chennai	21614.092031
Delhi	29461.983471
Hyderabad	20555.048387
Kolkata	11645.173664
Mumbai	85321.204733

dtype: float64

```
# plot of average rent by city
data.groupby('City')['Rent'].mean().plot(kind='bar')
```

<Axes: xlabel='City'>

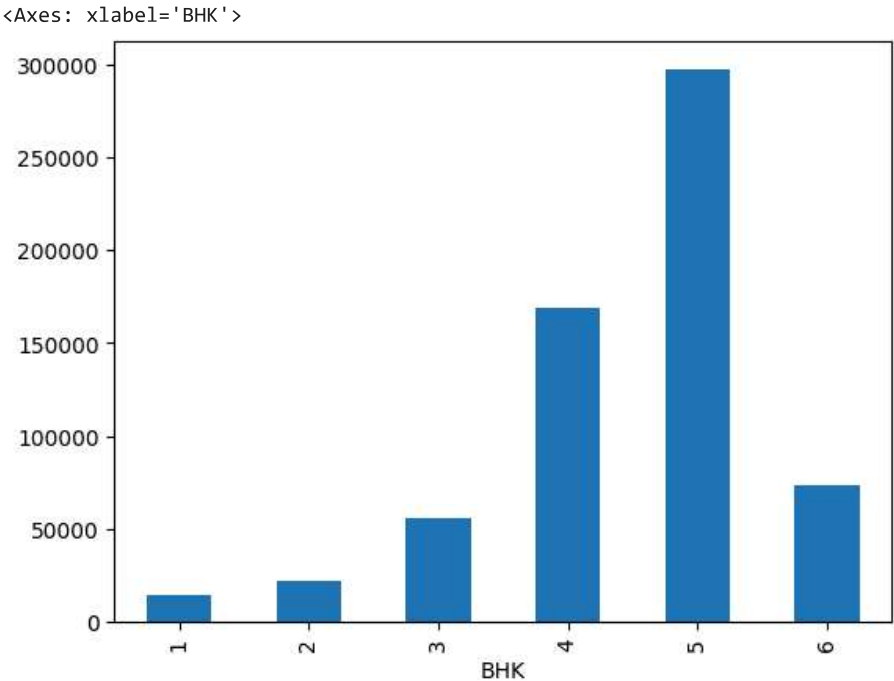


```
# average rent by number of bhk
data.groupby('BHK')['Rent'].mean()
```

Rent	
BHK	
1	14139.223650
2	22113.864018
3	55863.062842
4	168864.555556
5	297500.000000
6	73125.000000

dtype: float64

```
# plot of average rent by bhk
data.groupby('BHK')['Rent'].mean().plot(kind='bar')
```



```
# count of values by BHK
data['BHK'].value_counts().sort_index()
```

count	
BHK	
1	1167
2	2265
3	1098
4	189
5	19
6	8

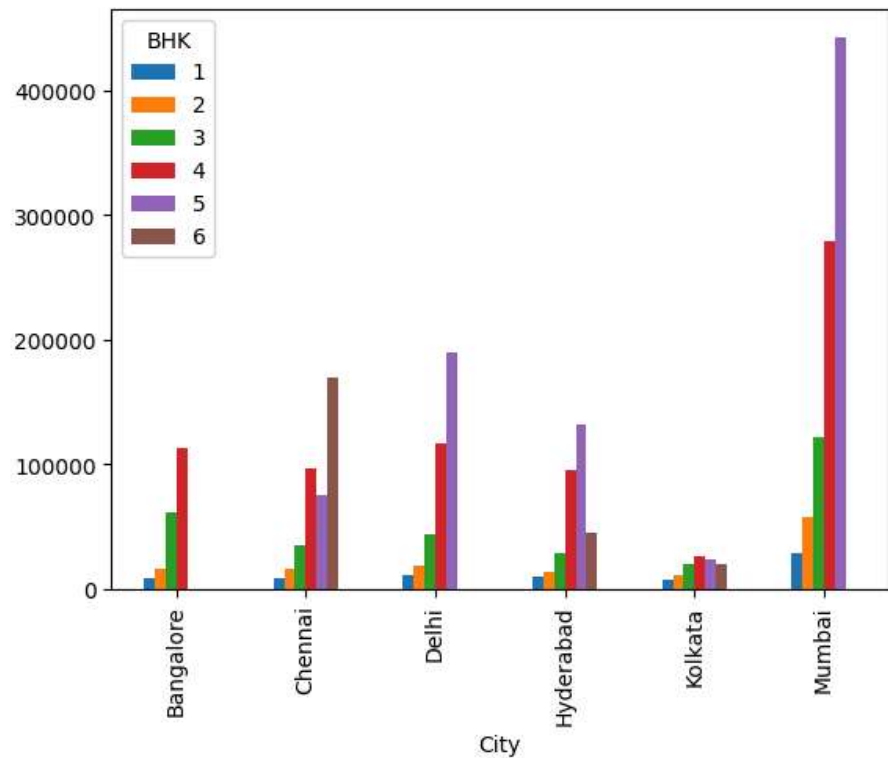
dtype: int64

```
# average rent by city & bhk
data.groupby(['City', 'BHK'])['Rent'].mean()
```


Rent		
City	BHK	
Bangalore	1	9368.333333
	2	16122.041763
	3	61988.580247
	4	113043.478261

```
# plot of average rent by city bhk
data.groupby(['City','BHK'])['Rent'].mean().unstack().plot(kind='bar')
```

```
<Axes: xlabel='City', ylabel='Rent'>
```



```
# unique values of area type
data['Area Type'].unique()
```

```
array(['Super Area', 'Carpet Area', 'Built Area'], dtype=object)
```

```
5 23750.000000
```

```
6 20000.000000
```

```
# create a map of area types & replace in data
area_map = {'Super Area':1, 'Carpet Area':2, 'Built Area':3}
data['Area Type'] = data['Area Type'].map(area_map)
data.head()
```

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